

To the Moon!

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Bob, having lost all of his money buying coffee, has recently decided to get rich quick by investing his entire life savings into a highly volatile, completely unproven cryptocurrency called MadesCoin.

For the past n days, he has been staring at the charts. He has recorded an array $a_1, a_2, \dots, a_n$ where a_i is the closing price of MadesCoin on day i .

Skeptical of her brother's life choices, Alice decides to analyse MadesCoin's performance and needs your help. She defines a Green Day as a day where the closing price of MadesCoin outperforms the previous day's price (day i is called a Green Day if $a_i > a_{i-1}$).

She gives you q queries. Each query consists of two integers: a starting day l and an ending day r . Your task is to tell her how many Green Days occurred during that period.

Important: When analysing a specific period from day l to day r , the very first day of that period (day l) can never be considered a Green Day, because its preceding day ($l-1$) falls outside the queried timeframe. You are only looking for price increases that happened entirely within the given window.

Input

The first line contains two integers n and q ($2 \leq n \leq 10^5$; $1 \leq q \leq 10^5$) — the number of days Bob tracked the price, and the number of queries Alice asked, respectively.

The second line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) — the closing price of MadesCoin on each day.

The following q lines each contain two integers l and r ($1 \leq l < r \leq n$) — the starting day and ending day of the queried period.

Output

For each query, output a single integer on a new line — the number of Green Days that occurred in the period from day l to day r inclusive.

Examples

standard input	standard output
8 3 3 1 4 1 5 9 2 6 1 8 3 6 2 3	4 2 1
4 1 10 10 10 10 1 4	0